

Katherine Mueller

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Education

Georgia Institute of Technology Online	Jan 2024 – Present
Master of Science in Computer Science (Specialization: Computer Graphics), GPA 3.67/4.00	
Harvard University Graduate School of Design Cambridge, MA	Sep 2022 – May 2025
Master of Design Studies (Domain: Publics), GPA not provided	
Georgia Institute of Technology Atlanta, GA	Aug 2017 – Dec 2021
Bachelor of Science in Industrial and Systems Engineering (Minor: Architecture), GPA 3.89/4.00	

Experience

Continuum , 80% Travel, 20% Remote <i>Industrial Engineering Project Specialist</i>	Jun 2025 – Oct 2025
- Developed engineered labor standards and conducted detailed time studies for three operational functions at DigiKey's one-million-sq-ft distribution center in Thief River Falls, MN, identifying process improvement opportunities - Validated existing labor standards and performance metrics for additional functions to support the rollout of the Takt labor-management system, ensuring accuracy and consistency in productivity benchmarks - Led cross-functional alignment and presented implementation updates to site leadership and floor supervisors, communicating technical findings and driving adoption of standardized work practices	
Chicago Trading Company , Chicago, IL <i>Quantitative Trading Intern</i>	Jun 2024 – Aug 2024
- Completed firm-led training in options theory and market-making fundamentals, including <i>Basics of Options</i> and <i>Intro to Options Market Making</i> - Assisted Global Risk Management in refining risk parameter models to improve consistency in options risk calibration and reporting across asset classes - Built Sigma dashboard to evaluate investment opportunities in international options markets, integrating Bloomberg Python API data with internal Snowflake databases	
Avangrid , Boston, MA <i>Data Engineering Intern – Energy Analytics</i>	May 2023 – Dec 2023
- Automated energy and pricing data retrieval using Python and Selenium WebDriver, streamlining quarterly reporting workflows - Converted Excel VBA system into a relational SQL-based database using Microsoft Access as the backend, enabling easier integration with Python scripts - Developed Power BI dashboard template for energy price forecasting visualizations	
Delta Air Lines , Atlanta, GA <i>International Network Planner – Caribbean Leisure Markets</i>	Jan 2022 – Jul 2022
- Designed post-pandemic service patterns for Caribbean leisure routes, aligning schedules with evolving demand and resource availability - Conducted rapid scheduling analyses to adjust Caribbean route plans when pilot block-hour constraints emerged, prioritizing high-demand markets during post-COVID network recovery - Developed scheduling plan and implementation timeline for international operations at Delta's new LAX terminal, assessing Federal Inspection Services (FIS) staffing feasibility and timing constraints	
SimTigrate Design Lab , Atlanta, GA <i>Undergraduate Research Assistant</i>	May 2021 – Dec 2021
- Reviewed and synthesized literature on evidence-based design strategies for mental and behavioral health environments - Analyzed and distilled insights from stakeholder interviews to inform spatial and operational requirements for new behavioral health facility proposals - Created patient journey maps illustrating experiences within 24-hour crisis care facilities to identify environmental stressors and design improvement opportunities	

East Atlanta Engineering for Kids, Atlanta, GA

Jan 2021 – May 2021

Teacher and Administrative Assistant

- Prepared and taught after-school engineering lessons for students in Pre-K through 5th grade, implementing interactive projects to introduce core STEM concepts
- Prepared lesson materials, supply lists, and schedules, and coordinated logistics and parent communication for upcoming summer programs

ExxonMobil, Houston, TX

Jun 2020 – Aug 2020

Global Supply Chain Intern

- Developed performance standards for third-party logistics (3PL) management, identifying process inefficiencies and improvement opportunities valued at up to \$1.7M annually
- Conducted cost analyses of packaging and multi-modal transportation options, recommending network optimizations projected to reduce logistics costs by \$244K per year

ExxonMobil, Baton Rouge, LA

Jan 2019 – May 2019

Process Engineering Intern – Crude Oil Distillation

- Scoped and budgeted a new piping-line proposal to improve process throughput with potential to reduce capital costs by \$6M
- Diagnosed and resolved heat-transfer-fluid flow imbalance in the air-preheater system; developed and implemented a test plan that improved thermal efficiency and reduced operating costs by \$250K per year
- Supported refinery turnaround activities by entering distillation towers to inspect trays and coordinating management-of-change (MOC) approvals across departments to ensure safe implementation of system modifications

Dow Chemical Company, Marlborough, MA

May 2018 – Jul 2018

Process Engineering Intern

- Reduced tank cycle time to increase production throughput, saving approximately \$25K per batch
- Developed standardized batch sheet for recycling ion-exchange canisters, improving documentation and process consistency
- Investigated source of missing solvent through cross-department interviews and documentation review, identifying reporting gaps and resolving material-tracking discrepancies

Skills**Programming:** Python, Java, C#, JavaScript, HTML, CSS, R, MATLAB, Git, Gurobi**Software & Tools:** Power BI, Microsoft Office Suite, Arduino IDE, Android Studio, VS Code, Simio**Design & Visualization:** Rhino 3D, Grasshopper, TouchDesigner, Adobe Creative Suite, OpenGL, SketchUp